AquaNexis -- Round 2 Progress Report

Track: 2 Water Purifier | Institution: UPES Dehradun | Participation: Solo Prototype | Date: 4 Nov 2026 | Competition: atomQUEST 2026

1. Project & Participant Details

Field	Value
Project	AquaNexis -- Predictive Water Intelligence System
Track	Track 2: Water Purifier
Institution	UPES Dehradun
Participation	Solo Prototype
Primary	Filter failure prognostics (FSI, FHI, RUL, Failure Risk)
Secondary	Beyond-TDS sensing (pH, Turbidity, Temp, Flow, Pressure)

2. Problem & Solution

Problem: Purifiers lack predictive intelligence. Replacements reactive/time-based (wasteful). No visibility into filter degradation, quality trends, impending failures.

Solution: Software-only platform ingesting 13-sensor telemetry, computing FSI/FHI via deterministic models, estimating RUL with confidence, classifying failure risk with evidence, providing real-time dashboard, alerts, guided maintenance. All readings/analytics carry explicit provenance (MEASURED/SIMULATED/CALCULATED/MODEL_PREDICTED).

3. Completed Software Modules

Module	Status	Key Features
Telemetry Ingestion	Done	FastAPI, JWT, dedup, quality validation
Water Quality Analytics	Done	AquaScore 0-100, param scores, trends, anomalies
Filter Stress Index (FSI)	Done	7-comp deterministic, pressure diff, rolling baselines
Filter Health Index (FHI)	Done	5-comp weighted, maintenance bands, confidence
RUL Estimation	Done	ML interface + deterministic fallback, stability, counterfactuals
Failure Risk	Done	5-comp risk model, explainable evidence, alert integration
Alert State Machine	Done	DETECTED->OPEN->ACK/SNOOZE/RESOLVE/AUTO_CLEAR
Notifications	Done	Console, Email/SMTP, Webhook with SSRF protection
Maintenance Lifecycle	Done	Catalog 11 cats, compat engine, workflow state machine
Calibration Mgmt	Done	Time/usage policies, post-maint verification, baselines
Safety Architecture	Done	Leak/pressure/flow/pump, idempotent, RBAC
Real-time WebSocket	Done	Telemetry/alert broadcast, exp backoff reconnect
React Dashboard	Done	Sensor grid, AquaScore, FHI, Diagnostics, Alerts, Maint
Provenance Display	Done	MEASURED/SIMULATED/CALCULATED/MODEL_PREDICTED badges

4. Original Sourcing Risks & Status

Risk	Status	Resolution
ESP32 hardware unavailable	Resolved	Software-only scope; firmware archived
ESP-IDF build env missing	Resolved	Not needed for software prototype
Physical sensor calibration	Resolved	Simulator provides calibrated synthetic data
Sensor PCB fabrication	Resolved	Out of scope - simulated telemetry
Custom filter housing	Resolved	Out of scope - catalog only

5. Budget: Original vs Actual

Category	Estimate	Actual	Notes
ESP32 Dev Kit	1,500	0	Not purchased - out of scope
Sensors (13 types)	8,000	0	Not purchased - out of scope
PCB Fabrication	2,000	0	Not purchased - out of scope
Enclosure/Mechanical	3,000	0	Not purchased - out of scope
Cloud/Hosting	500	0	Local dev only
Total	~15,000	0	Software-only - no hardware

Note: Stage 1 estimated Rs.5,000-10,000 for minimal hardware. Actual Rs.0 - pivoted to software-only before procurement.

6. Scope Change & Reason

Original (Stage 1): HW+SW purifier with ESP32, 13 physical sensors, custom PCB, filter housing, dashboard.

Revised (G6): Software-only. Python simulator replaces sensors. All analytics, dashboard, alerts, maintenance, safety in software.

Reason: ESP-IDF unavailable; no hardware for validation; competition allows SW-only; focus on predictive/diagnostic value.

Firmware in firmware/esp32/ archived (13 drivers, comm layer, main app - GCC validated).

7. Revised Targets & Verification

Original Target	Revised Target	Status
>=4 contextual signals	13 sensors implemented	Verified - all 13 in sim & dashboard
RUL MAE <= 7 days	Deterministic fallback	Not verifiable - no trained model; labeled
Dashboard <= 5s update	WS + 5s staleTime	Verified - WS live, stale at 30s
Auto safe shutdown	SW safety alerts only	Verified - alerts generated; no actuators
TDS rejection/flow	Purification module	Verified - rejection, recovery, efficiency

8. Delay Reasons

Delay	Cause	Mitigation
ESP-IDF setup failed	No ESP-IDF/Docker on Windows	Pivot to SW-only; firmware archived
PostgreSQL not running	No local DB instance	Unit tests use mocks; 133/133 pass (11 DB errs)
WS message type mismatch	Backend/frontend types diverged	Mapping layer in WS manager
Offline mode missing	No connection state handling	Stale banner, offline indicator, auto-recovery

9. Safety / Fail-Safe Status

Feature	Implementation	Status
Leak detection	Reading -> critical alert -> safety workflow	SW implemented
High pressure	Threshold breach -> alert + pump shutdown cmd	SW command only
Low flow/dry run	Flow < threshold -> alert + shutdown cmd	SW command only
Pump overcurrent	Current > threshold -> alert + shutdown cmd	SW command only
Physical actuator	NOT IMPLEMENTED - SW prototype	EXPLICITLY NOT DONE

Critical: No physical purifier, pump, valve, or actuator exists. All safety actions are SW alerts/logged commands. System demonstrates fail-safe logic but cannot physically shut down equipment.

10. Round 3 Plan

Activity	Description	Target
Ext Telemetry API	Documented API for real purifier ingest (HTTP/WS)	Q1 2027
ML Model Training	Collect data -> train RUL -> replace fallback	Q2 2027
Mobile App	React Native for technicians/consumers	Q2 2027
Fleet View	Dashboard for multiple purifiers	Q3 2027
Field Validation	Partner with OEM for real-world pilot	Q3 2027
Regulatory	Map analytics to ISO/NSF standards	Q4 2027

11. Demo Evidence (G7.6)

Scenario	Evidence
NORMAL	Online, AquaScore 85, sensors GOOD, prov=SIMULATED
FILTER_STRESS	FSI=65 MOD, FHI=45 POOR, Band=REPLACE, Alert=Critical
LEAK	Critical "Leak Detected", Diags=CRITICAL, safety workflow
SENSOR_FAULT	Quality=SUSPECT/BAD, provenance tracked, no silent degradation

All tested via python backend/simulator.py --profile <name>.

Declaration

Reflects software-only implementation. All HW dev (ESP32, sensors, PCB, enclosure) out of scope, archived in firmware/esp32/. Demo uses simulated telemetry from backend/simulator.py. No physical purifier exists.

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